Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

SELADEVS UTHM

Selected Domain

Domain 1 - AI, Future Skills & Workforce

Selected SDG Track (1–17)

SDG 4 primary; SDGs 8 and 10 secondary

Project / Solution Name

SHIFTATLAS 3D

Name (Leader)

MUHAMMAD HAZIQ FARHAN BIN NORDIN

Email & Phone (Leader)

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University Affiliation(s)

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

Number of Team Members

4 registered members

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

We are building SHIFTATLAS 3D for Malaysian tertiary learners and early-career workers who need a clearer route from their existing experience to relevant skills during AI-driven change. Occupational descriptions, research signals and learning resources are scattered, while many course lists give little explanation of what to learn first. We want learners to leave with a skill profile they can review, a role comparison they can understand and an affordable plan they can act on. Hays' 2025 Malaysia technology/AI release reports that 56% of surveyed professionals used AI at work while only 26% received training. We use this as survey context rather than a population estimate or evidence of demand for a particular role. Our approach brings these decisions into connected 3D worlds, backed by official state-level labour statistics and traceable references. We distinguish state aggregates from institution coordinates and individual career guidance. SDG 4 is our main contribution through access to skills and lifelong learning, with related pathways to SDGs 8 and 10. We have not yet measured changes in employment, salary, proficiency or training access. Source: Hays Malaysia, 2025, Future of Work: Malaysia technology and AI impact.

Hays Malaysia (2025): https://www.hays.com.my/press-release/content/future-of-work-malaysia-technology-ai-impact

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We built SHIFTATLAS 3D around a practical journey: understand our experience, explore suitable roles and choose what to learn next. Learners review ten self-assessed skills, compare six O*NET-referenced roles and follow free-first learning plans ordered by prerequisites, hours and content budget. Our local TF-IDF/cosine retrieval links fictional experience to role descriptions, while an authored weighted rubric explains each score and gap. Dated research-title signals can adjust existing weights by at most 25%; we present them as topic signals rather than hiring forecasts. Resume parsing stays in the browser, with keyword excerpts reviewed before any skill mention is accepted at a minimum level of 1. We connected profiles, trends, careers, learning, all 17 SDGs and two uncertain adoption scenarios through 3D worlds with text equivalents. Our Malaysia map combines DOSM boundaries and 2026 Q2 statistics for all 16 states/federal territories; institution and landmark pins have separate coverage. Atlas Quest rewards exploration with activity badges. Our cloud extension combines multilingual E5 relevance with Career Council: one Qwen3-4B-Instruct-2507 model assigned four roles - profile reviewer, career explorer, learning planner and critic. Each stage builds on earlier suggestions, and the critic challenges the reasoning to support comparison. Our server validates known identifiers, exact profile quotations and prerequisites; it preserves self-assessed levels and rubric scores. The same model supplies all Council viewpoints, so its critic may share its errors. Our published Replit gateway completed a six-role E5 CUDA request with dated research context. We also recorded Qwen profile-stage CUDA execution. Our website and both cloud configuration routes are publicly reachable. Our demonstration uses fictional inputs, keeps resume files local and requires explicit cloud consent. The result is an inspectable route from experience to a practical learning plan, with sources and uncertainty visible throughout.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

We built our application with Next.js 16.4.2, React/DOM 19.3.1 and TypeScript 5.9.3. Our 3D worlds use Three.js 0.180.0, React Three Fiber 9.8.2 and Drei 10.7.9. We use PDF.js 6.4.299 and Mammoth 1.13.0 for local document extraction, with Node/npm for our server and build.

Our flow connects public snapshots to profile review, local TF-IDF and weighted role comparison, then ordered learning and uncertain scenarios. We present each step through 3D and text views. Our optional Node gateway uses trusted catalogues to connect E5 relevance or four Council reviews without changing the local rubric.

We pinned multilingual-e5-base (MIT) and Qwen3-4B-Instruct-2507 (Apache-2.0). We built E5 and completed synthetic CUDA requests for E5 and Qwen profile review. Our public Next interface and both API status routes respond. The Replit E5 gateway completed a six-role CUDA request. Council uses its own endpoint. We use sponsor credit only. Workshop learning informed responsible GenAI, structured data and deployment; our prototype uses versioned snapshots and local state rather than a backend database.

Live application: https://sdc-2026-seladevsuthm.replit.app/
Repository: https://github.com/hzqfarhan/SDC2026-SELADEVSUTHM
YouTube demo: https://youtu.be/e8jtdJy6deo

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We shaped SHIFTATLAS around one question: can a learner turn existing experience into a next step they understand and can afford? Through prototype iteration and interface feedback, we refined the light-theme navigation, source disclosures, inspectable 3D layers and evidence-review flow. We separated profile review, role comparison and learning priorities, and added text controls and reduced motion for different devices. Our work so far is a functioning prototype rather than a completed user study. For Phase 2, we plan to recruit ten learners, compare plan-completion time against a static resource list and check whether they can explain a score contribution and scenario uncertainty. We will collect baseline timings, task failures and misleading results before estimating improvement. We support iteration with versioned snapshots, refresh scripts, hashes and automated checks. We published our interface on Replit and ran synthetic E5 and Qwen profile-review CUDA jobs on RunPod. These steps establish a working foundation for our next evaluation: multilingual relevance, explanation quality and practical learning choices. Our goal is decision support that learners can inspect, challenge and revise, with an affordable route from exploration to action.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Impact measurement plan

We will evaluate whether SHIFTATLAS helps learners turn exploration into a practical plan. Our Phase 2 pilot will recruit ten learners to compare plan-completion time against a static resource list, then test understanding of a score contribution and both scenario uncertainties.

We will collect baseline timings, task accuracy, failed tasks and misleading results. Our targets below are evaluation goals; we have not yet measured learner outcomes or national employment effects.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	We prioritise SDG 4.3 and 4.4 through affordable learning access and relevant skills. Our related pathways support SDG 8.2 / 8.5 and 10.2 through productive skills and inclusion. We mapped all 17 goals to relevant targets and distinguish direct from indirect connections. These mappings express our intended contribution rather than measured impact or UN endorsement.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	We draw context from Hays Malaysia (2025): 56% of surveyed professionals used AI at work, while 26% received training, a 30-percentage-point gap within that survey. Our own evaluation starts with ten planned learners. We will measure baseline plan-completion time before calculating improvement; we have not yet collected pilot outcomes.

Hays Malaysia (2025): https://www.hays.com.my/press-release/content/future-of-work-malaysia-technology-ai-impact

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	We will track median minutes to produce an actionable plan, correct explanations of a role-score contribution and both scenario uncertainties, and counts of failed tasks or misleading results. Our planned sample is ten learners. We will assess decision quality and usability rather than infer placement or salary effects.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	We aim for at least 20% lower median plan-completion time than the same participants using a static resource list, and at least 80% correct explanation checks. We will test these targets after collecting baseline data and record recruitment, task order and completion criteria. This small pilot will guide our next iteration.

Our evaluation is planned. We have not measured resource savings, energy savings or causal SDG effects.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [ ] All fields in this template have been completed within the word limits stated.

All four of us completed the three mandatory workshops. We built a functional prototype grounded in public sources, published our Replit interface and recorded real CUDA requests.

We are completing added-cloud review and final acceptance steps before certifying full readiness.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

We began our project work after 19 October 2026, 20:00 MYT. Our submission combines our project concept and implementation with separately declared frameworks, public data and pretrained models. We do not claim those third-party materials as our original work.

We used OpenAI Codex extensively during development. All four of us reviewed the core prototype and guide and can explain its major components; we are completing our review of the added cloud code. We published our Replit interface and completed synthetic E5 and Qwen profile-review CUDA jobs on RunPod. Our public E5 gateway completed a six-role CUDA request; complete Council acceptance is separate.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next/React; Three/R3F/Drei	Code/frameworks	MIT
PDF.js; TypeScript; Mammoth	Code/tools	Apache-2.0; BSD-2-Clause
DOSM; Wikidata; MQA facts	Public data	CC BY 4.0; CC0; source terms
UN / arXiv / O*NET; E5 / Qwen	Assets/refs; E5 and Qwen-profile CUDA fixtures	UN/source terms; MIT / Apache-2.0
Synthetic profiles; authored rubric	Original project data	Project Apache-2.0

We ran E5 and Qwen profile-stage CUDA fixtures. Sources, revisions and notices are in our declaration and repository; full Council verification is separate.

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

None of us has an employment, internship or consulting agreement that gives an employer ownership of this project.

We have not included employer-owned code, confidential datasets or workplace credentials. We retain the applicable licences and attribution for our third-party materials.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

Signature

_____

Name: MUHAMMAD HAZIQ FARHAN BIN NORDIN

[Date]

25 October 2026